Mathematics for Computing

Assignment 1

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Sets

Set

A Set is a well-defined, unordered collection of disticnt elements

Cardinality of a Set

The number of elements in a set.

Example:
A data scientist chooses the following machine learning algorithms.

$$A = \{\text{Linear Regression}, \text{Decision Tree}, \text{Random Forest}, \text{SVM}, \text{XGBoost}\}$$
$$|A| = 5$$

Types of Sets

Finite Set

A set with a limited number of elements.

Example:

A sentiment analysis dataset has only three labels.

$$L = \{\text{Positive}, \text{Neutral}, \text{Negative}\}$$

Types of Sets

Infinite Set

A set with unlimited elements.

Example:

A neural network can receive infinitely many real-valued inputs.

$$X = \{x \mid x \in [0, 1]\}$$

Types of Sets

Null (Empty) Set

A set with no elements.

Example:
After filtering a dataset for patients older than 200 years, no records remain.

$$P = \{x \mid x > 200\} = \varnothing$$

Types of Sets

Equal Sets

Two sets containing exactly the same elements.

Example:

Two ML engineers list identical preprocessing steps.

$$A = \{\text{Scaling}, \text{Imputation}, \text{One-Hot Encoding}\}$$
$$B = \{\text{One-Hot Encoding}, \text{Scaling}, \text{Imputation}\}$$
$$A = B$$

Types of Sets

Equivalent Sets

Two sets having the same number of elements.

Example:
Three ML models and three deep learning frameworks.

$$A = \{\text{SVM}, \text{Decision Tree}, \text{Random Forest}\}$$
$$B = \{\text{TensorFlow}, \text{PyTorch}, \text{Keras}\}$$
$$|A| = |B| = 3$$

Types of Sets

Subset

Every element of one set belongs to another set.

Example:
Here set A consists of the Deep Learning models and set B consists of the Machine learning models
Deep learning models are a subset of machine learning algorithms.

$$A = \{\text{CNN}, \text{RNN}, \text{Transformer}\}$$
$$B = \{\text{CNN}, \text{RNN}, \text{Transformer}, \text{SVM}, \text{Decision Tree}\}$$
$$A \subseteq B$$

Types of Sets

Superset

A set containing all elements of another set.

Example:

Here set A consists of the Deep Learning models and set B consists of the Machine learning models

Machine learning algorithms include all deep learning models.

$$A = \{\text{CNN}, \text{RNN}, \text{Transformer}\}$$

$$B = \{\text{CNN}, \text{RNN}, \text{Transformer}, \text{SVM}, \text{Decision Tree}\}$$

$$B \supseteq A$$

Types of Sets

Universal Set

A set containing every element under consideration.

Example:
All AI algorithms studied in the course.

$$U = \{\text{Linear Regression}, \text{Decision Tree}, \text{Random Forest}, \text{SVM}, \text{CNN}, \text{RNN}, \text{Transformer}\}$$

Types of Sets

Power Set

The set of all possible subsets.

Example:

Feature selection using Age and Income.

$$A = \{\text{Age}, \text{Income}\}$$
$$\mathcal{P}(A) = \{\varnothing, \{\text{Age}\}, \{\text{Income}\}, \{\text{Age}, \text{Income}\}\}$$

Types of Sets

Singleton Set

A set containing exactly one element.

Example:

Hyperparameter tuning selects only Adam.

$$O = \{\text{Adam}\}$$

Types of Sets

Countable Set

A set whose elements can be counted one by one.

Example:

Datasets stored with unique IDs.

$$D = \{D_1, D_2, D_3, \ldots\}$$

Union

The union of two sets contains all elements that belong to either set or both.

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

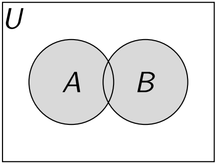
Example:

Let

$$A = \{\text{CNN}, \text{RNN}\}, \qquad B = \{\text{RNN}, \text{Transformer}\}$$

Then,

$$A \cup B = \{\text{CNN}, \text{RNN}, \text{Transformer}\}$$



Operations on Sets

Intersection

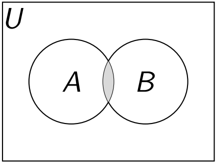
The intersection of two sets contains the common elements.

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

Example:

$$A = \{\text{CNN}, \text{RNN}\}, \quad B = \{\text{RNN}, \text{Transformer}\}$$

$$A \cap B = \{\text{RNN}\}$$



Operations on Sets

Difference of Sets

The difference of sets contains elements of A that are not in B.

$$A - B = \{x \mid x \in A,\ x \notin B\}$$

Example:

$$A = \{\text{CNN}, \text{RNN}\}, \quad B = \{\text{RNN}, \text{Transformer}\}$$

$$A - B = \{\text{CNN}\}$$

Set Difference

The set difference $B - A$ contains elements of B that are not in A.

$$B - A = \{x \mid x \in B,\ x \notin A\}$$

Example:

$$B - A = \{\text{Transformer}\}$$

Symmetric Difference

The symmetric difference contains elements that belong to either set but not both.

$$A \triangle B = (A - B) \cup (B - A)$$

Example:

$$A = \{\text{CNN}, \text{RNN}\}$$

$$B = \{\text{RNN}, \text{Transformer}\}$$

$$A \triangle B = \{\text{CNN}, \text{Transformer}\}$$

Complement

The complement of A contains all elements of the universal set that are not in A.

$$A' = U - A$$

Example:

$$U = \{\text{CNN}, \text{RNN}, \text{Transformer}\}$$

$$A = \{\text{CNN}, \text{RNN}\}$$

$$A' = \{\text{Transformer}\}$$

Operations on Sets

Cartesian Product

The Cartesian product of two sets is the set of all ordered pairs.

$$A \times B = \{(a, b) \mid a \in A,\ \ b \in B\}$$

Example:
Let

$$A = \{\text{Image}, \text{Text}\}$$

$$B = \{\text{CNN}, \text{Transformer}\}$$

Then,

$$A \times B = \{(\text{Image}, \text{CNN}), (\text{Image}, \text{Transformer}), (\text{Text}, \text{CNN}), (\text{Text}, \text{Transforme}$$

Examples of Sets in Set Builder form

1.Squares of first 10 natural numbers

$$S = \{n^2 \mid n \in \mathbb{N},\ 1 \leq n \leq 10\}$$

2. Set of even numbers in range of 1 to 10

$$E = \{x \in \mathbb{N} \mid 1 \leq x \leq 10,\ x \text{ is even}\}$$

Examples of Sets in Set Builder form

3.Set of prime numbers in less than 20

$$P = \{x \in \mathbb{N} \mid x < 20,\ x \text{ is prime}\}$$

4. Set of all integers between -10 and 10

$$I = \{x \in \mathbb{Z} \mid -10 \leq x \leq 10\}$$

Examples of Sets in Set Builder form

5.Set of all natural numbers which are multiples of 3 but not 7

$$A = \{x \in \mathbb{N} \mid 3 \mid x,\ 7 \nmid x\}$$

6.K=$\{$5,9,7,13,17,....$\}$

$$K = \{4n + 1 \mid n \in \mathbb{N}\}$$

7.Set of all natural numbers leaves 2 as remainder when divided by 6

$$A = \{x \in \mathbb{N} \mid x = 6n + 2,\ n \in \mathbb{N}_0\}$$

8.A=$\{$7,14,21,28,...$\}$

$$A = \{7n \mid n \in \mathbb{N}\}$$

9.Set of factors of 26

$$F = \{x \in \mathbb{N} \mid x \mid 26\}$$

10.S=$\{$(4,2),(4,3),(5,2),(8,5),(3,2),(6,4)$\}$

$$S = \{(x, y) \mid x, y \in \mathbb{N},\ x > y\}$$

Inclusion-Exclusion Principle

The Inclusion-Exclusion Principle is used to find the number of distinct elements in overlapping sets by adding the sizes of the sets and subtracting their common elements to avoid double counting.

For two sets,

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example:

Let

- A = Users interested in AI ($|A| = 120$)
- B = Users interested in Data Science ($|B| = 100$)
- $|A \cap B| = 30$

Then,

$$|A \cup B| = 120 + 100 - 30 = 190$$

Hence, **190 unique users** are interested in AI or Data Science.

General Form of Inclusion-Exclusion Principle

For Two Sets:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

For Three Sets

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

General Formula (for n sets)

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} \left|\bigcap_{i=1}^{n} A_i\right|$$

Partially Ordered Sets (Posets)

Partially Ordered Sets

A **Partially Ordered Set (Poset)** is a set together with a binary relation $\preceq$ that satisfies the following properties for all $a, b, c \in P$:

- **Reflexive:** $a \preceq a$
- **Antisymmetric:** If $a \preceq b$ and $b \preceq a$, then $a = b$
- **Transitive:** If $a \preceq b$ and $b \preceq c$, then $a \preceq c$

A poset allows some elements to be incomparable, meaning not every pair of elements must have an order.

Example: Consider the set of machine learning model complexities,

$$P = \{\text{Linear Regression},\ \text{Decision Tree},\ \text{Random Forest},\ \text{Deep Neural Network}\}$$

Partially Ordered Sets (Posets)

Define the relation $\preceq$ as "*is less complex than or equal to.*" Then,

$$\text{Linear Regression} \preceq \text{Decision Tree} \preceq \text{Random Forest} \preceq \text{Deep Neural Network}$$

This forms a partially ordered set because the relation is reflexive, antisymmetric, and transitive. In practice, such ordering helps compare models based on their complexity, making it useful in AI model selection and Data Science workflows.

Example for POSET

Example1: Image Processing
Let

$$P = \{\text{Image Acquisition},\ \text{Image Enhancement},\ \text{Image Segmentation}\}$$

Define the relation $\preceq$ as "*performed before.*"

$$\text{Image Acquisition} \preceq \text{Image Enhancement}$$

while **Image Segmentation** is independent of **Image Enhancement** for some applications. Hence, $(P, \preceq)$ is a **partially ordered set** because not every pair of operations is comparable.

Examples for POSET

Example2: Deep Learning Models Let

$$P = \{\text{Data Preprocessing},\ \text{Model Training},\ \text{Model Evaluation}\}$$

Define the relation $\preceq$ as "*performed before*."

Then,
Data Preprocessing $\preceq$ Model Training and Model Training $\preceq$ Model Evaluation.

Hence, $(P, \preceq)$ is a **partially ordered set** because the tasks follow a dependency order.

Hasse Diagram

Hasse Diagram

A **Hasse Diagram** is a graphical representation of a partially ordered set (poset). It shows only the **covering relations** by omitting reflexive, transitive, and redundant edges.

Construction Rules

- Place smaller elements below larger elements.
- Connect only directly related (covering) elements.
- Do not draw arrowheads.
- Omit reflexive and transitive relations.

Hasse Diagram

Let a relation R be defined on a set A

$$A = \{a, b, c\}$$

$$R = \{(a, a), (b, b), (c, c), (d, d), (a, b), (a, c), (b, d), (c, d), (a, d)\}$$

Removing the reflexive relations,we get

$$R = \{(a, b), (a, c), (b, d), (c, d), (a, d)\}$$

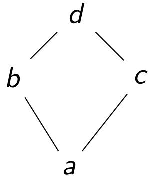
Since,

$$a \prec b, \quad b \prec d \;\Rightarrow\; a \prec d$$

the transitive relation (a, d) is omitted in the Hasse diagram.
Then we get the relation

$$R = \{(a, b), (a, c), (b, d), (c, d)\}$$

Hasse Diagram



$$a \prec b, \quad a \prec c, \quad b \prec d, \quad c \prec d$$

Example of Hasse Diagram

Example: Deep Learning Workflow

Let,

$P = \{\text{DataCollection}, \text{DataPreprocessing}, \text{ModelTraining}, \text{ModelEvaluation}\}$

with the relation "performed before".

$$\text{DataCollection} \prec \text{DataPreprocessing} \prec \text{ModelTraining} \prec \text{ModelEvaluation}$$

$$\begin{tikzcd}
\text{Model Evaluation} \arrow[d, dash] \\
\text{Model Training} \arrow[d, dash] \\
\text{Data Preprocessing} \arrow[d, dash] \\
\text{Data Collection}
\end{tikzcd}$$

Thank You!